

UNIT – 3
ASSESSMENT – 1
REPORT

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1. Objective:

To implement a rule-based medical diagnosis system using Forward Chaining and Backward Chaining.

Methodology

1. Store patient symptoms as facts.
2. Define diagnosis rules.
3. Apply Forward Chaining to derive possible diagnoses.
4. Select Pneumonia as the target goal.
5. Apply Backward Chaining to find the conditions required for Pneumonia.
6. Verify whether those conditions exist in the knowledge base.

Result

Forward Chaining derives:

Possible Flu

Possible Pneumonia

Backward Chaining successfully verifies:

Patient A has Pneumonia

2. Objective:

To implement an FOL-based traffic management agent using unification and resolution.

Methodology

1. Define the FOL rules.
2. Identify emergency vehicle facts.
3. Apply unification to substitute variables.
4. Derive ClearPath.
5. Derive GreenSignal.
6. Apply Rule 3 to derive Proceed.
7. Negate the goal.
8. Use resolution to derive the empty clause.

Result

The system successfully proves:

The KB can support multiple emergency vehicles if the corresponding vehicle and emergency facts are provided.

3. Objective:

To convert an agricultural knowledge base into CNF and prove ApplyDripMethod using Resolution Refutation.

Methodology

1. Define the agricultural rules.
2. Convert implications into CNF.
3. Add known facts.
4. Negate the goal.
5. Apply resolution.
6. Derive the Empty Clause.
7. Analyze the effect of removing SoilDry.

Result:

With SoilDry present:

is proved.

After removing SoilDry, neither ApplyDripMethod nor CropAtRisk can be logically established.